

ECON 101—Introduction to Microeconomics
Spring 2026

MIDTERM II

March 26, 2026

General Instructions

- The exam has a duration of **65 minutes**. If you finish early, you may leave within the first 45 minutes only. Otherwise remain quietly until the exam concludes.
- All answers must be written in the answer sheet. Scratch work will not be considered. You may not take this booklet with you.
- **Allowed on your desk or person:** ID, writing instruments, ruler, and a simple calculator. **Nothing else.** No extra paper, no graphing calculators, no phones, no smart devices, no electronics, no unauthorized materials of any kind. Any item not explicitly listed above must be placed outside the room or at the front before the exam begins. No exceptions.
- You may request a bathroom break, but only one person may be out at a time.
- Crying is allowed, but quietly, and not on the exam paper.
- Good luck and may your choice of inputs maximize your profits!

Exam Instructions and Academic Dishonesty Policy

Before the exam begins:

- **Allowed on your desk or person:** ID, writing instruments, ruler, and a simple calculator. **If it is not on this list, it is not allowed.** All prohibited items must be placed outside the room or at the front before the exam begins.
- Wait until everyone is seated. Do **not** write anything—including your name and ID—until you are explicitly told to do so.

At the end of the exam:

- When time is called, put all writing instruments down immediately, close your booklet, keep your hands away from it, and remain seated until dismissed.

Academic Dishonesty:

- Any form of cheating, including any deviation from the instructions above, will be met with the **harshest consequences available**. There will be **no sympathy and no leniency**.

Answer Sheet Instructions

- **Record your exam version on the answer sheet.** If no version is indicated, the lowest-scoring version will be applied. No exceptions.
- Use only the provided pencil and erase cleanly if you make a mistake.
- Examples:
 - **Correct:** Q.1. ☒ ☐ (B) ☐ (C) ☐ (D)
 - **Incorrect (cross):** Q.2. ☒ (X) ☐ (B) ☐ (C) ☐ (D)
 - **Incorrect (two bubbles):** Q.3. ☒ ☒ ☐ (C) ☐ (D)
- This exam has 20 questions worth a total of 100 points plus one bonus question worth 3 points. Point values are indicated in parentheses and reflect perceived difficulty—by me, though you may think otherwise.

Exam Version 3

This exam has 20 questions, worth 3, 5 or 7 points depending on perceived difficulty, for a total of 100 points. You may disagree. There is also a bonus question worth 3 points.

The following information pertains to questions 1–5. Daniyar fancies himself a true warrior of the steppe—though his battlefield is the highway from Astana to Semey, 750km away. Instead of a car, he has two horses. His noble steed Tulpur is fast but hungry, galloping at 50 km/h and consuming 2 kg of hay per kilometer. His second horse, Boz, is slower but frugal, trotting at 20 km/h and consuming only 0.5 kg of hay per kilometer. He can switch between horses freely at any time with no cost. Each kg of hay costs 100 tenge. Denoting x_1 and x_2 as the distances traveled, in kilometers, on Tulpur and Boz respectively, Daniyar strictly prefers any bundle (x_1, x_2) such that $x_1 + x_2 = 750$ to any other bundle, and is indifferent between any two such bundles. He wants to spend no more than I tenge and arrive in no more than H hours.

1. (5 points) Which of the following is an appropriate utility representation of Daniyar's preferences?

- (a) $U(x_1, x_2) = x_1 + x_2$
- (b) $U(x_1, x_2) = -(x_1 + x_2 - 750)^2$
- (c) $U(x_1, x_2) = x_1^{375} x_2^{375}$
- (d) $U(x_1, x_2) = \min\{x_1, x_2\}$
- (e) I too wish I were a warrior of the steppe right now.

2. (5 points) Which of the following correctly describes the constraints of Daniyar's optimization problem? Consider the following:

- I. $\frac{x_1}{50} + \frac{x_2}{20} \leq H$
 - II. $200x_1 + 50x_2 \leq I$
 - III. $x_1, x_2 \geq 0$
- (a) II and III only
 - (b) I and II only
 - (c) I identify with Daniyar: lost, no hay, no horses.
 - (d) I, II, and III.
 - (e) I and III only

3. (7 points) Now suppose $H = 27$. What is the minimum level of income I required for Daniyar to reach Semey?

- (a) 150,000 tenge
- (b) 37,500 tenge
- (c) I too am trying to reach a destination and failing.
- (d) 90,000 tenge
- (e) 75,000 tenge

4. (7 points) Given your answer above, what is the optimal bundle (x_1, x_2) that Daniyar will choose?

- (a) (750, 0)
- (b) I am also 750km away from understanding this problem.
- (c) (400, 350)
- (d) (350, 400)
- (e) (0, 750)

5. (7 points) Now suppose $I = 150,000$ tenge. Which of the following statements is correct?
- (a) The solution is unique at (350, 400)
 - (b) There are multiple optimal bundles (x_1, x_2) that allow Daniyar to reach Semey
 - (c) At this point I would also like 150,000 tenge.
 - (d) The solution is unique and Daniyar rides Tulpar the entire way
 - (e) The solution is unique and Daniyar rides Boz the entire way

The following information pertains to questions 6–9. Almira likes sweet pastries s and coffee c for breakfast. Her preferences can be represented by:

$$U(c, s) = 3 \log c + 2 \log s$$

The price of coffee is p_c , the price of pastries is p_s , and her income is m .

6. (3 points) What is Almira's marginal rate of substitution MRS_{cs} at the bundle $(c, s) = (2, 2)$?
- (a) $3/2$
 - (b) I have optimized my time away from studying quite effectively.
 - (c) $2/3$
 - (d) 2
 - (e) 1
7. (3 points) What fraction of her income does Almira spend on coffee at her optimal bundle?
- (a) $1/2$
 - (b) I maximize utility by not solving maximization problems.
 - (c) $3/2$
 - (d) $2/5$
 - (e) $3/5$
8. (3 points) Which of the following correctly states Almira's demand for coffee and pastries?
- (a) $c^* = \frac{m}{5p_c}, s^* = \frac{m}{5p_s}$
 - (b) I am at a corner solution: zero studying, maximum regret
 - (c) $c^* = \frac{3m}{2p_c}, s^* = \frac{2m}{3p_s}$
 - (d) $c^* = \frac{3m}{5p_c}, s^* = \frac{2m}{5p_s}$
 - (e) $c^* = \frac{2m}{5p_c}, s^* = \frac{3m}{5p_s}$
9. (3 points) Are coffee and pastries normal or inferior goods?
- (a) Coffee is normal, pastries are inferior
 - (b) My grade is about to be inferior, does that count?
 - (c) Both are inferior goods
 - (d) Coffee is inferior, pastries are normal
 - (e) Both are normal goods

10. (5 points) Consider the utility function:

$$u(x_1, x_2) = \begin{cases} x_1 + x_2 & \text{if } x_1 \leq 1, \\ -x_1 + x_2 + 2 & \text{if } x_1 > 1. \end{cases}$$

Which of the following properties of consumer preferences does this utility function violate?

- I. Monotonicity
 - II. Transitivity
 - III. Completeness
- (a) III only
 - (b) I only
 - (c) I and II, but not III
 - (d) II only
 - (e) All of the above

11. (5 points) Sultan has a utility function given by

$$u(x_1, x_2) = \sqrt{x_1} + x_2$$

Both prices are equal to 1 and his income satisfies $I > \frac{1}{4}$. If his income increases, which of the following statements is true?

- (a) Quantity demanded of both x_1 and x_2 stays the same
- (b) Quantity demanded of x_1 stays the same, quantity demanded of x_2 increases
- (c) Club activities were more important to me, and that is why I have no clue.
- (d) Quantity demanded of both x_1 and x_2 increases at the same rate
- (e) Quantity demanded of x_1 increases, quantity demanded of x_2 doesn't change

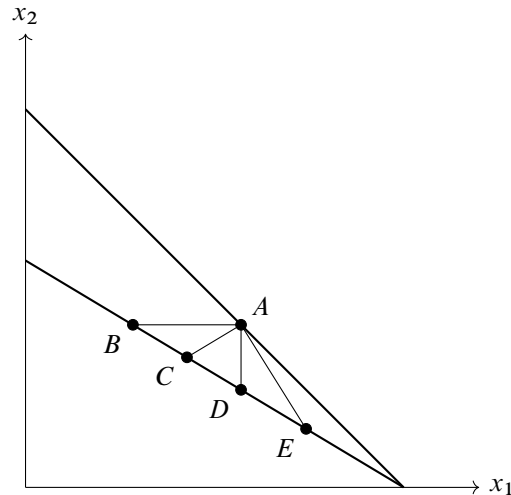
12. (7 points) Now suppose $p_1 = \frac{1}{16}$, $p_2 = 1$, and $I = 2$. If Sultan's income increases to $I = 3$, which of the following statements is true?

- (a) Quantity demanded of x_1 stays the same, quantity demanded of x_2 increases
- (b) Quantity demanded of x_1 increases, quantity demanded of x_2 doesn't change
- (c) I have no idea, but I am very passionate about extracurriculars.
- (d) Quantity demanded of both x_1 and x_2 increases at the same rate
- (e) Quantity demanded of both x_1 and x_2 stays the same

13. (7 points) Now assume $p_1 = p_2 = 1$ and $I = 10$. Which of the following statements is correct regarding the effect of a price change in good 1 on quantity demanded of x_1 ?

- (a) There is no income effect; the change is purely a substitution effect
- (b) The income effect cannot be determined without more information
- (c) There is a positive income effect; x_1 is a normal good
- (d) I was going to attend office hours, but I was busy not studying.
- (e) There is a negative income effect; x_1 is an inferior good

Based on the graph presented below, answer questions 14–15



14. (5 points) Suppose Daryn spends all his income consuming goods x_1 and x_2 and is initially consuming bundle A . If there is an increase in the price of x_2 and the demand for x_2 is inelastic, Daryn's new equilibrium point will be _____ and the goods are _____.
- C ; complements
 - E ; independent
 - I prefer not to answer on the grounds that I may incriminate myself.
 - E ; substitutes
 - B ; substitutes
15. (7 points) Suppose Daryn spends all his income consuming goods x_1 and x_2 and is initially consuming bundle A . In which of the following cases would Daryn decide to consume bundle D ?
- An increase in the price of x_2 and the demand for x_2 is perfectly inelastic.
 - An increase in the price of x_2 and the demand for x_2 is elastic.
 - An increase in the price of x_2 and the demand for x_1 is inelastic.
 - None of the above, none of the below, none of anything I studied.
 - An increase in the price of x_2 and the demand for x_2 is unit elastic.
16. (5 points) Nurai orders a coffee x_1 and a cookie x_2 from Daily Cup, where x_1 and x_2 represent the quality (taste) of each item. Her enjoyment depends only on the better quality of the two items. Which of the following properties of consumer preferences do these preferences violate?
- Convexity
 - Monotonicity
 - Completeness
- II only
 - III only
 - I and III, but not II
 - I only
 - All of the above

17. (3 bonus points) What is the name of Prof. Alejandro's cat?
- (a) Gordolfo
 - (b) I did not study this either.
 - (c) Peluso
 - (d) Panqueque
 - (e) Bigotes
18. (7 points) Aruzhan consumes only two goods, x_1 and x_2 . When the price of x_1 increases, she consumes less of good x_2 . Consider the following statements:
- I. The substitution effect of an increase in p_1 on x_2 is positive.
 - II. The income effect of an increase in p_1 on x_2 is negative.
 - III. x_2 is a normal good.
 - IV. x_2 cannot be a Giffen good.
 - V. x_1 is an inferior good.
- Which of the following is correct?
- (a) I, II and III only
 - (b) I, II, III and IV only
 - (c) II, III and IV only
 - (d) I and IV only
 - (e) All of the above
19. (3 points) Mansur takes his coffee very seriously: he simply refuses to eat a croissant without two espressos to wash it down — no more, no less. If the price of espresso increases, which of the following statements is correct?
- (a) Demand for both espressos and croissants decreases, entirely due to the substitution effect.
 - (b) Demand for espresso decreases but demand for croissants increases, due to the substitution effect.
 - (c) I also enjoy coffee and pastries, but not this question.
 - (d) Demand for espresso decreases but demand for croissants stays the same, entirely due to the income effect.
 - (e) Demand for both espressos and croissants decreases, entirely due to the income effect.
20. (3 points) Malika has an income of 100 tenge. She consumes only two goods: pineapples (x_1) and takis (x_2), where quantities are discrete (whole units only). Her utility function is:

$$u(x_1, x_2) = x_1 x_2$$

The prices are 20 tenge per pineapple and 50 tenge per taki. However, the only shop where takis are sold offers a special discount: for every third pineapple, she gets it for free. Which consumption bundle (x_1, x_2) maximizes Malika's utility under this discount?

- (a) (5, 0)
- (b) (0, 2)
- (c) (2, 1)
- (d) (3, 1)
- (e) I should have studied more, but here we are.

21. (3 points) How does Malika's utility with the discount compare to her utility without it?

- (a) Higher than without discount
- (b) Same as without discount
- (c) Lower than without discount
- (d) I should have come to class. I know.
- (e) Cannot be compared